

ECON-7102: Environmental Economics I

Class 4: Theory of Environmental Policy

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Fall 2026

Today's Agenda

1. Questions?
2. Theory of environmental policy
 - Damage functions and abatement costs
 - Property rights and Coase
 - Environmental policy instruments
3. For next class...

Environmental policy

- Environmental policy seeks to limit the effects of human activity on the environment
- Economists seek *efficient* solutions to manage pollution
 - Efficient solutions set optimal pollution levels where marginal benefits of pollution abatement equal marginal costs
 - Need to know the benefits of pollution abatement (in dollars).
 - We can think of (the negative of) these benefits as pollution *damages*
- For now, we're assuming:
 - Full information
 - Pollution and output are separable
 - **Flow** pollutants (like air pollution); not stock pollutants (like GHG emissions)
 - Fully competitive output markets
 - No geographic/temporal dimensions

Damage functions and abatement costs

- Polluting firms, indexed by $j = 1, \dots, J$
 - Firms produce emissions $e_j \implies$ total emissions: $E = \sum_{j=1}^J e_j$
 - Pollution is “well-mixed” and affects all households equally

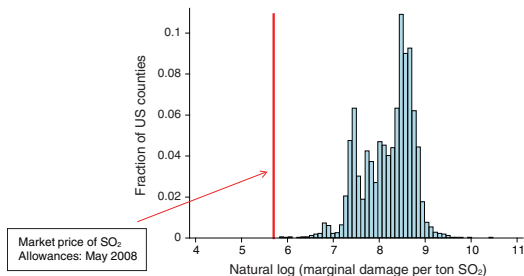
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- Households (i) have utility: $U_i(y_i, E) = y_i - D_i(E)$
 - y_i is income
 - $D_i(E)$ is the dollar value of lost utility from aggregate emissions
 - Assume D_i is increasing and weakly convex, i.e., $D'_i > 0$, $D''_i \geq 0$

Damage functions and abatement costs

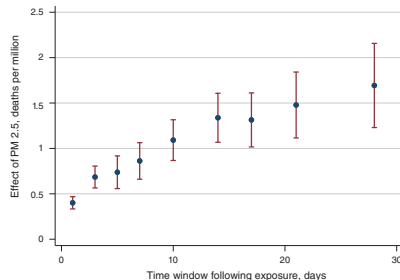
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 - Assume D_i is increasing and weakly convex, i.e., $D'_i > 0$, $D''_i \geq 0$
- Total damages are given by: $D(E) = \sum_{i=1}^I D_i(E)$

Damage functions in practice



Simulated damages: disperse one ton of SO₂, add up downwind exposure, monetize the effects. Damages are nearly all premature mortality via sulfate PM_{2.5}, so $D'(E)$ depends population exposure.

Source: Muller, Mendelsohn, and Nordhaus. "Environmental Accounting for Pollution in the United States Economy." AER, 2011.



Estimated damages: wind direction increases PM_{2.5} exposure $\Rightarrow$ $\uparrow$ deaths. One day of $+1 \mu\text{g}/\text{m}^3$ raises elderly mortality by 0.69 per million over three days. Causal, but acute mortality only.

Source: Deryugina, Heutel, Miller, Molitor, and Reif. "The Mortality and Medical Costs of Air Pollution." AER, 2019.

Abatement costs

- $C_j(e_j)$ is firm j 's cost of abatement (i.e., reducing emissions)
- If firms' emissions are greater than private optimum ($e_j > \hat{e}_j$), then $C_j(e_j) = 0$
- If $e_j < \hat{e}_j$
 - $C_j(e_j) > 0$ (costs are positive)
 - $C'_j(e_j) < 0$ (marginal reductions in emissions marginally increase costs)
 - Define $MAC_j(e_j) = -C'_j(e_j) > 0$
 - where $MAC'_j(e_j) = -C''_j(e_j) \leq 0$
 - Marginal abatement costs are decreasing in emissions (i.e., increasing in abatement).
 - A little abatement is cheap, but becomes more costly as abatement increases.

Damages and abatement costs: **Efficiency**

- Pareto optimality $\implies$ a trade-off between negative impact of externality on utility and the benefit of emissions as an input to production.
- Society's objective: minimize social costs (i.e., abatement costs + pollution damages)

$$\min_{e_j} SC(e_1, \dots, e_J) = \sum_{j=1}^J C_j(e_j) + D(E)$$

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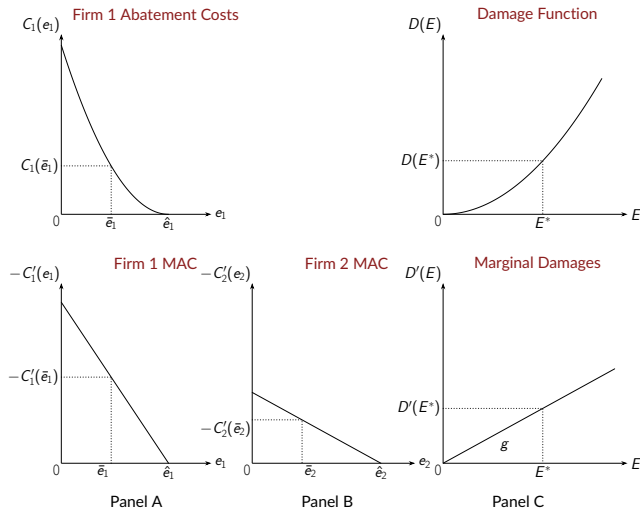
$$\min_{e_j} SC(e_1, \dots, e_J) = \sum_{j=1}^J C_j(e_j) + D(E)$$

$$\implies -C'_j(e_j) = D'(E) \quad \forall j$$

$$\implies -C'_j(e_j) = -C'_k(e_k) \quad \forall j, j \neq k$$

- **Optimal emissions occur where marginal abatement costs = marginal damages**
- **Marginal abatement costs are equalized across all firms**

Social cost minimization

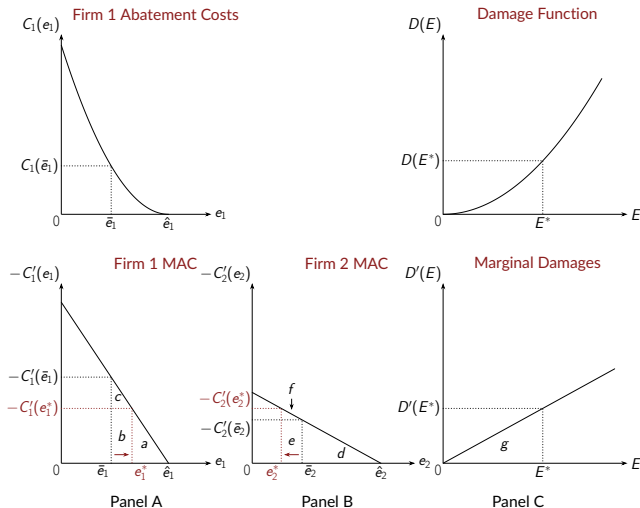


Damages functions and abatement costs

- Abatement costs are decreasing in emissions.
- $MAC = -C'$ also decreasing in emissions
- E^* is optimal total emissions
 - Where $-C'_j = D'$
 - Firm 1 emits e_1^*
 - Firm 2 emits e_2^*
 - $MAC_1(e_1^*) = MAC_2(e_2^*)$
- Area under MAC curve is total cost of achieving abatement level E^*

Figure: Pollution damages, costs, and efficient emission level

Social cost minimization



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Figure: Pollution damages, costs, and efficient emission level

The Coase Theorem

- In the case of an externality...
- If...
 - Property rights are clearly defined,
 - There are minimal transactions costs,
 - There aren't too many parties involved,
- Then...
 - The efficient solution can be achieved through bargaining *regardless* of who is given property rights.

Decentralized bargaining can achieve socially optimal level of emissions irrespective of who has property rights. Each agent doesn't want to compensate other party for more than it's worth at the margin.

The Coase Theorem

Discussion

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- Should polluters pay for their damages?
- What is the role of government in Coasean bargaining?
- What types of environmental problems are likely to be solved through Coasean bargaining?
- Coase vs. Pigou

Indeed, Pigou's treatment of the problems considered in this article is extremely elusive and the discussion of his views raises almost insuperable difficulties of interpretation. Consequently it is impossible to be sure that one has understood what Pigou really meant. Nevertheless, it is difficult to resist the conclusion, extraordinary though this may be in an economist of Pigou's stature, that the main source of this obscurity is that Pigou had not thought his position through.

IX. THE PIGOVIAN TRADITION

It is strange that a doctrine as faulty as that developed by Pigou should have been so influential, although part of its success has probably been due to the lack of clarity in the exposition. Not being clear, it was never clearly wrong. Curiously enough, this obscurity in the source has not prevented the emergence of a fairly well-defined oral tradition. What economists think they learn from Pigou, and what they tell their students, which I term the Pigovian tradition, is reasonably clear. I propose to show the inadequacy of this Pigovian tradition by demonstrating that both the analysis and the policy conclusions which it supports are incorrect.

I do not propose to justify my view as to the prevailing opinion by copious references to the literature. I do this partly because the treatment in the literature is usually so fragmentary, often involving little more than a reference to Pigou plus some explanatory comment, that detailed examination

Coasean bargaining: A simple household-firm model

HH has right to clean environment

- Firm offers an emissions/payment contract (E, θ) for right to pollute.
- Household will accept if $\theta \geq D(E)$

Firm minimizes total costs:

$$\min_E C(E) + \theta + tr$$

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$$\implies -C' = D'$$

- Firm will offer $\theta = D(E)$ to compensate for damages.
- Requires transaction costs $<$ gains from negotiation.

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- Requires transaction costs $<$ gains from negotiation.

Firm has right to pollute

- Household offers an emissions/payment contract (E, θ) for firm to limit emissions.
- Firm will accept if $\theta \geq C(E)$

Household maximizes utility:

$$\max_E \{y - [D(E) + \theta + tr]\}$$

$$\max_E \{y - [D(E) + C(E) + tr]\}$$

$$\implies -C' = D'$$

- Household will offer $\theta = C(E)$ to compensate for abatement costs.
- Requires transaction costs $<$ gains from negotiation.

Outside of private (i.e., Coasean) negotiation, we have several tools to reduce pollution:

- Command and control (i.e., emissions standards)
- Emissions taxes
- Pollution permits:
 - Auctioned permits
 - Freely distributed, transferable permits
- Subsidies

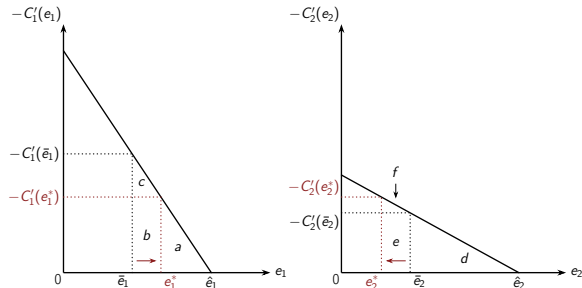
Command and control

- Government (i) sets limits on quantity of pollution coming from each firm or (ii) mandates that they use a specific technology.
- Ideally, government sets e_j^* for each firm, where e_j^* is optimal emissions.
 - Requires knowledge of social damage function AND individual firms' abatement curves.
- Instead, regulator sets a **uniform emissions standard**: $\bar{e} = E^*/J$, where E^* is socially optimal level of aggregate emissions.

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- Instead, regulator sets a **uniform emissions standard**: $\bar{e} = E^*/J$, where E^* is socially optimal level of aggregate emissions.

- If firms are identical, then $\bar{e} = e_j^* \forall j$
- If firms are heterogeneous, policy will be more costly than efficient solution.
- $-C'_1(\bar{e}) \neq -C'_2(\bar{e})$
- Marginal costs are not equalized across firms.
- Cheaper for Firm 2 to abate more than Firm 1.
- Potential for trading: Firm 1 should be willing to pay Firm 2 to abate.



Emissions taxes

- Following Pigou, impose a tax on polluting firms' emissions.
- Firms minimize total costs:

$$\min_{e_j} TC_j(e_j) = C_j(e_j) + \tau e_j \quad \implies -C'_j(e_j) = \tau$$

- Firms emit up to the point where MAC equals the tax rate.
- $MAC_j(e_j) = \tau = MAC_k(e_k)$ for $j \neq k$ (marginal costs are equalized across firms)
- MAC s are decreasing in τ

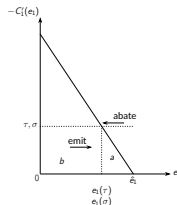
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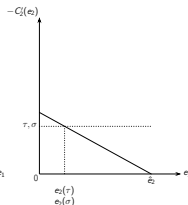
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- Firms emit up to the point where *MAC* equals the tax rate.
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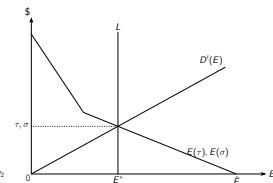
- How to choose τ ?
 - Need to know all *MAC* curves to achieve $e_j^*(\tau)$.
- Taxes also raise revenue.
 - What the government does with that revenue matters for policy.



Panel A



Panel B



Panel C

Auctioned pollution permits

- Firms must purchase permits for the “right to pollute” (e.g., 1 permit = 1 ton of pollution).
 - Offer L pollution permits, which corresponds to the optimal level of emissions, E^* .
 - An auction price σ emerges from competitive bidding by firms.
- Firms still minimize total costs:

$$\min_{e_j} C_j(e_j) + \sigma e_j \quad \implies \quad -C'_j(e_j) = \sigma$$

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- Firms will emit up to point where marginal abatement costs = equilibrium permit price
 - If $\sigma > MAC_j(e_j)$, firm will abate emissions.
 - If $\sigma < MAC_j(e_j)$, firm will purchase emissions permits.
- Similar to taxes, government/regulator earns revenue from auctioning permits

Equivalence of taxes and permits

Proposition 3.2 (P&R)

If the regulator knows the polluting industry's aggregate demand curve for emissions, or equivalently the aggregate marginal abatement cost curve, and if the firms behave as price takers on the permit market, the regulator can achieve an aggregate emissions target $\bar{E}$ by:

1. Issuing a total amount of permits for auction $L = \bar{E}$; or
2. Charging an emissions tax τ which leads to $\bar{E}$. In this case, $\tau(\bar{E}) = \sigma(L)$ where $\sigma(L)$ is the market clearing price in the auction.

Furthermore, with knowledge of the damage function $D(E)$ the regulator can implement the socially optimal emission level E^* using either instrument.

Transferable (tradeable) permits

- Practically speaking, many cap-and-trade systems allocate permits based on historical emissions.
 - Regulator chooses total number of permits, L , and distributes them according to some rule.
 - Firms then can engage in trade, effectively creating a market for permits.
- Firm j is endowed with $\bar{e}_j$ permits; σ is the market price of permits.

$$\min_{e_j} C_j(e_j) + \sigma(e_j - \bar{e}_j) \quad \implies \quad -C'_j(e_j) = \sigma$$

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- Same solution as auctioned permit case (endowment is lump-sum transfer)
 - If $MAC_j(e_j) < \sigma$, then firm will emit up until $MAC_j(e_j) = \sigma$ or sell permits.
 - If $MAC_j(e_j) > \sigma$, then firm will purchase permits until $MAC_j(e_j) = \sigma$
 - Demand/supply for permits and demand/supply for emissions reaches an equilibrium where MACs are equalized across firms (and equal to σ).

Transferable (tradeable) permits

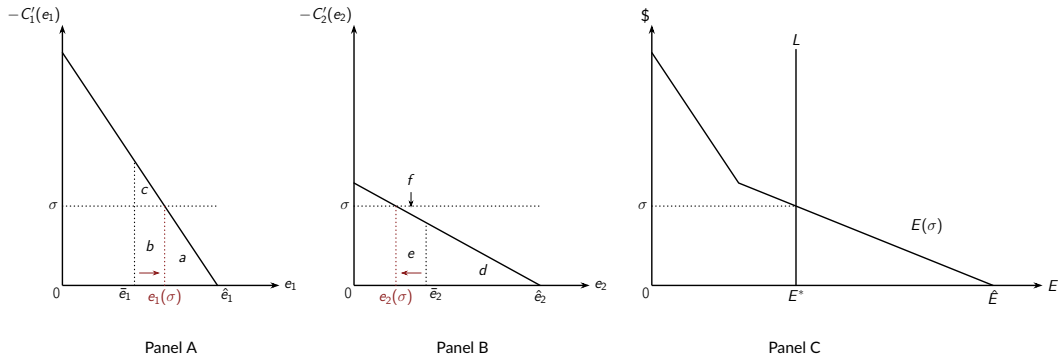


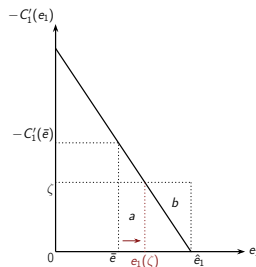
Figure: Freely distributed transferable permits (P&R Fig 3.3)

Subsidized abatement

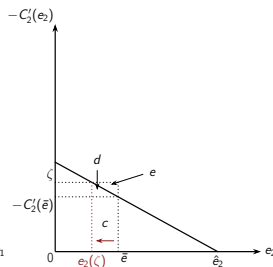
- Rather than requiring firms to bear the cost of abatement, governments can subsidize the provision of public goods (often for political economy reasons)
- Government provides a subsidy ζ for each unit of abatement below some reference level ($\bar{e}_j$).
- Firms minimize costs:

$$\min_{e_j} C_j(e_j) + \zeta(e_j - \bar{e}_j) \quad \Rightarrow \quad -C'_j(e_j) = \zeta$$

- Subsidies induce identical behavior at the margin as emissions taxes.
- *MACs* are equalized across firms
- Can achieve optimal emissions (E^*) if damage functions are known.
- Subsidies need to be financed from somewhere (taxes/auctions generate revenue)



Panel A



Panel B

Final thoughts

- Coasean bargaining can lead to efficient outcomes without government intervention
 - In practice, this is rare.
- Taxes, permits, subsidies can all achieve similar outcomes (in terms of emissions).
 - Often times, choice of policy instruments are driven by political considerations.
 - Each requires different information on abatement costs.
 - Standards are only efficient if firms' abatement costs are identical.
- Revenue/cost considerations matter:
 - Taxes and auctioned permits generate revenue, which can be used to make policies more efficient
 - Transferable permits result in intra-industry transfers
 - Subsidies need to be financed from governmental revenues

For next week...

caseyjwichman.com/econ7102/#week3

- **Monday:**

- No class (*Labor Day*)

- **Wednesday:**

- P&R, Ch. 4 (skim Sections 4.2 and 4.3)
- Weitzman (1974): “Prices vs. Quantities,” *Review of Economic Studies*, 1974.
 - **Reading response due for Weitzman.** Due: Tuesday, 11:59 PM. Submit via Canvas.